

RUNSHI TANG

rtang56@wisc.edu ◊ Madison, WI ◊ January 25, 2025

EDUCATION

Bachelor of Science

2016-2020

School of Mathematics, Shandong University

-Academic Excellence Award in 2016-2017, 2017-2018

(Exchanged to UW-Madison for further education in advance since 2019)

Master in Statistics - Data Science Track

2019-2021

Department of Statistics, University of Wisconsin-Madison

-Academic Excellence Award in 2019-2020

Ph.D. in Statistics

2021-Present

Department of Statistics, University of Wisconsin-Madison; Advised by Profs. Anru Zhang (Duke) and Richard Chappell (UW-Madison)

PAPERS

(†: equal contribution)

1. **Tang, R.**, Yuan, M., & Zhang, A. R. (2023). Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model. *Journal of the Royal Statistical Society, Series B*, qkae088.
2. Towe, S. L., **Tang, R.**, Gibson, M. J., Zhang, A. R., & Meade, C. S. (2023). Longitudinal changes in neurocognitive performance related to drug use intensity in a sample of persons with and without HIV who use illicit stimulants. *Drug and alcohol dependence*, 110923.
3. Zhang, A. R.[†], Bell, R.[†], An, C.[†], **Tang R.**[†], Shana Hall, S., Chan, C., Al-Khalil, K., & Meade, C. (2023). Cocaine Use Prediction with Tensor-based Machine Learning on Multimodal MRI Connectome Data. *Neural Computation*, 36.1 (2023): 107-127.
4. **Tang, R.**, Kolda, T., & Zhang, A. R. (2025+). Tensor Decomposition with Unaligned Observations. *Submitted to SIAM Journal on Matrix Analysis and Applications under Major Revision*.
5. Ko, A., Berger, M., Brown, D., Pac, J., **Tang, R.** (2025+). Unemployment Insurance Generosity and Child Protective Services Involvement *Working paper*
6. Pac, J., Berger, M., Brown, D., **Tang, R.** (2025+). Does Public Program Participation Impact Child Protective Services Involvement? Quasi-Experimental Evidence from the Staggered eWIC EBT Rollout. *Working paper*
7. Pac, J., Berger, L., Durrance, C., Nobles, J., and **Tang, R.** (2025) Foster care in the First Year of Life: Effects on Health and Safety. *Working paper*
8. **Tang, R.** et al. (2025+). Theoretical guarantee for ALS in Tensor CP Decomposition. *Working paper*

AWARDS

1. Honorable Mention Award in the 2024 Student Paper Competition of the ASA Statistical Learning and Data Science Section, “Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model.”
2. Student and Early Career Travel Fund by ASA
3. The SRGC Conference Presentation Award by the University of Wisconsin-Madison

TALKS AND PRESENTATIONS

1. The 18th annual CFAR Fall Scientific Retreat, “Longitudinal changes in neurocognitive performance related to drug use intensity in a sample of persons with and without HIV who use illicit stimulants. ”
2. The 37th New England Statistics Symposium, “Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model.”
3. 2024 Joint Statistical Meetings, “Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model.”

CONFERENCE/WORKSHOP ABSTRACTS

1. McCann, J., Bihlmeyer, N., Yang, C., Truong, T., **Tang, R.** et al. (2024). Unique Age and Sex Dependent Serum Metabolite Signature in Teens With Obesity. *Abstracts from the 42nd annual meeting of the Obesity Society at Obesity Week November 3-6, 2024*
2. Pac, J., Berger, L., Durrance, C., Nobles, J., and **Tang, R.** (2025) Foster care in the First Year of Life: Effects on Health and Safety. *2024 Association for Public Policy Analysis and Management and 2025 Atlanta Workshop on Public Policy and Child Well-being*

EXPERIENCES

Project Assistant

Sep 2022 - Now

Institute for Research on Poverty, UW-Madison

Madison, WI

- Supervised by Prof. Jessica Pac and Prof. Lawrence Berger (Institute for Research on Poverty & School of Social Science, UW-Madison)
- Study the relationship between child well-being and various policies
- Part of our results were presented by Prof. Lawrence Berger in the 2023 International Society for the Prevention of Child Abuse & Neglect Congress

Academia Internship

Summer, 2022

School of Medicine, Duke University

Durham, NC

- Supervised by Prof. Anru Zhang (Department of Biostatistics & Bioinformatics) and Prof. Christina Meade (Department of Psychiatry and Behavioral Sciences)
- Participated in two projects: longitudinal study on HIV contributes to neurocognitive impairments in persons who use stimulants and applying the novel tensor-based method on MRI data to predict cocaine use
- Presented a poster at the 18th annual CFAR Fall Scientific Retreat

Teaching Assistant

Sep 2021 - May 2022

Department of Statistics, UW-Madison

Madison, WI

- Fall 2021: STAT 324, Introductory Applied Statistics for Engineers
- Spring 2022: STAT 324, Introductory Applied Statistics for Engineers

Project Assistant

Sep 2020 - Aug. 2021

Department of Statistics, UW-Madison

Madison, WI

- Supervised by Prof. Awad Hanna (Department of Civil & Environmental Engineering, UW-Madison) and Prof. Wei-Yin Loh (Department of Statistics, UW-Madison)
- Using statistics and programming, Process and analyze data for a research project on construction material transportation of the Department of Civil & Engineering.

Actuarial Internship

Summer, 2018

Taishan Property Insurance Co., Ltd.

Jinan, Shandong Province, China

- Work for Actuarial Dept.; Learn structure of insurance company and knowledge of Property Insurance and making Actuarial Report; Help make observation report on opponents.

RESEARCH INTEREST

- High dimensional statistics, spiked covariance model, tensor decomposition, matrix analysis, optimization ...
- Applying theories into different practical fields, including social science and clinical science.